

ICA0002: IT Infrastructure Services

Git Basics

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Git

Distributed version control system (VCS)

Probably the most widely used VCS today

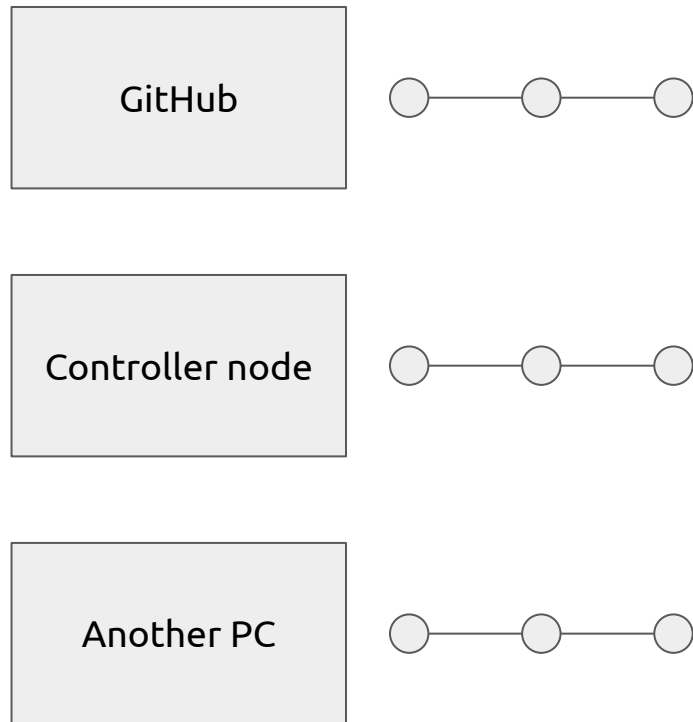
GitHub: service to host Git repositories

Git: distributed VCS

Multiple copies of the same code history

Pros:

- Your dog won't eat your homework -- at least not entirely



Git: distributed VCS

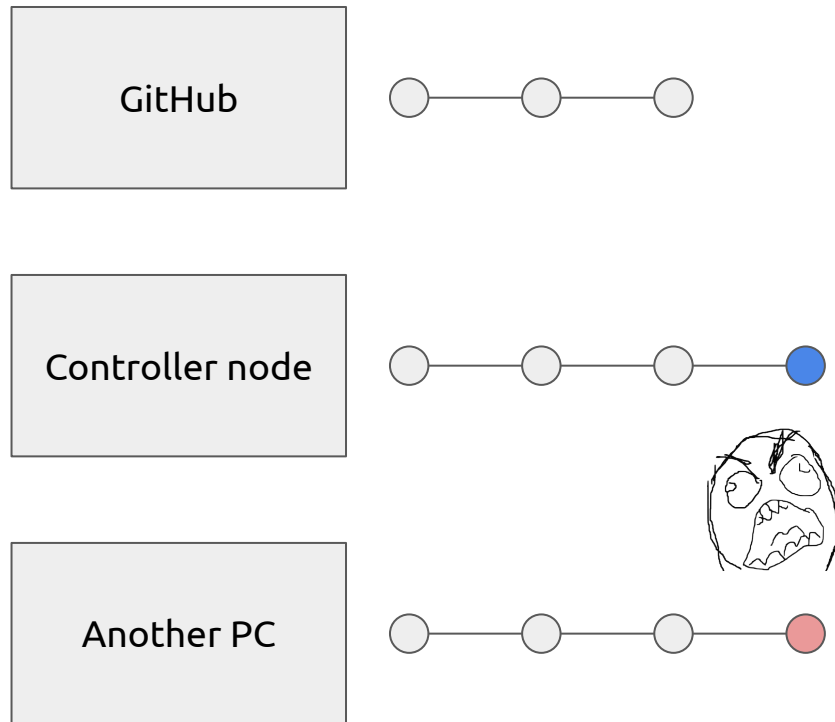
Multiple copies of the same code history

Pros:

- Your dog won't eat your homework -- at least not entirely

Cons:

- Management overhead -- Git is not as simple as **cp/mv**



Reminder

There is a Git GUI tool from GitHub: <https://github.com/apps/desktop>

You can use that if you want

- Not directly encouraged, but also not forbidden

Basic Git commands

`git init`

`git status`

`git add`

`git commit`

`git push`

`git pull`

`git clone`

Basic Git commands: init

`git init` Initialize new Git repository in this directory

Creates `.git` directory where repository info is stored

You need to do it only once during this course (if at all)

If you run it twice on different machines (even with the same files) you will initialize two **different** Git repositories

Basic Git commands: status, add, commit

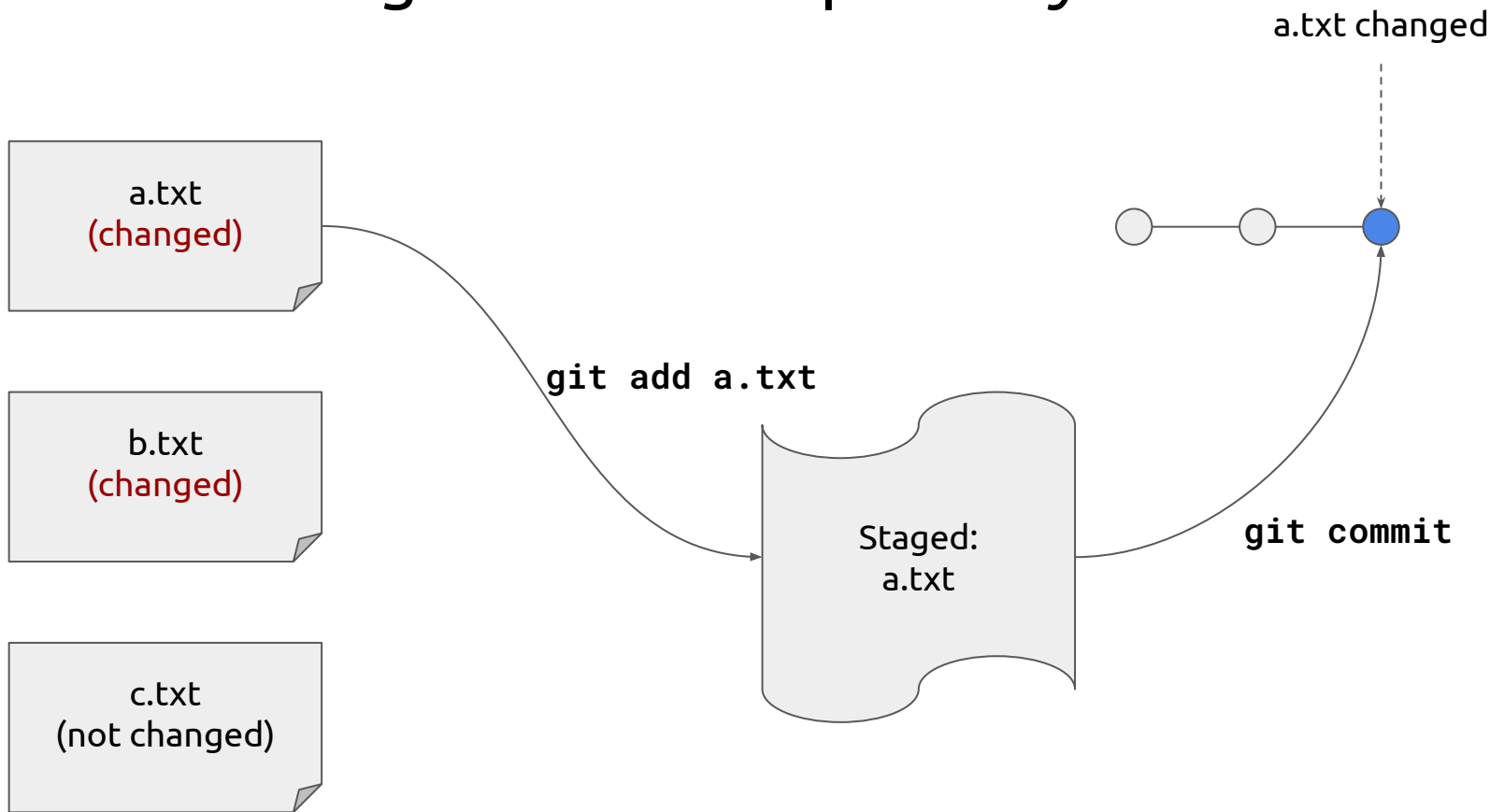
git status Show the changes and status of Git repository in this directory

git add . Mark all files in this directory as changed (stage)
Instead of '.' you may list individual files to stage

git commit Commit the staged files to the local Git repository

Note: committing the changes is a two step process (add + commit)

Git: commit changes to local repository



Basic Git commands: push, pull, clone

git push Push (upload) changes from local repository to Git server

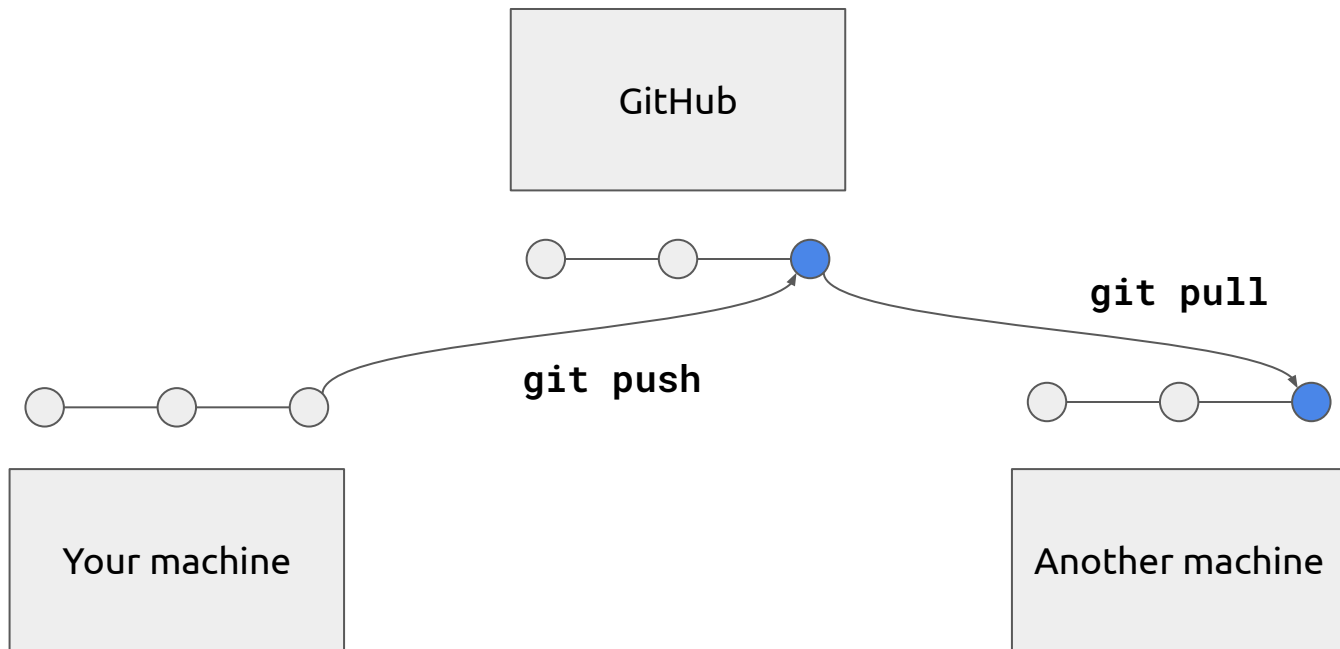
git pull Pull (download) changes from Git server to local repository

git clone git@github.com:<user>/<repo>.git

Clone entire repository from Git server

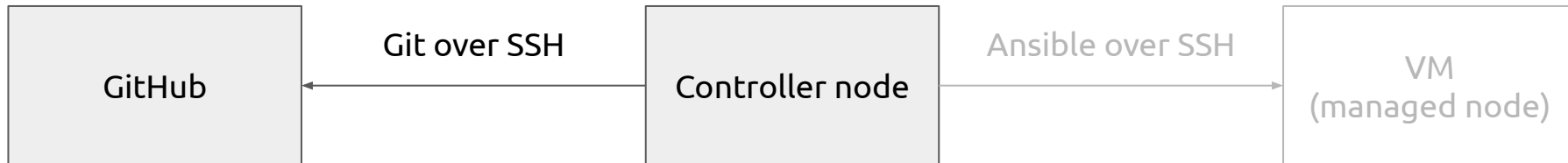
Expects that there is no local repository, and creates it

Basic Git commands: push, pull



Git + SSH

Git (and GitHub) support SSH authentication



Once your public SSH key is added to your GitHub account you can push, pull changes and clone repositories using only your SSH keys, without typing any passwords -- awesome!

Git usage in this course

Git is a bit overkill VCS for this course, but still it's the best tool we can use

A few recommendations if you still feel uncomfortable with Git:

1. If unsure, use one branch only: for example, **main**
2. **Always** pull changes from Git server before making changes locally
3. If you use more than one computer, make code changes on one at a time
4. Commit and push often -- it's free :)

These are solely for your convenience -- we are not checking if you follow them

In case of fire



1. git commit



2. git push



3. leave building

Questions?